

5. Build and Fire

A weekend to build, 48 hours to cure

Cut, glue, cure, test, shoot

THIS IS A PROJECTILE WEAPON

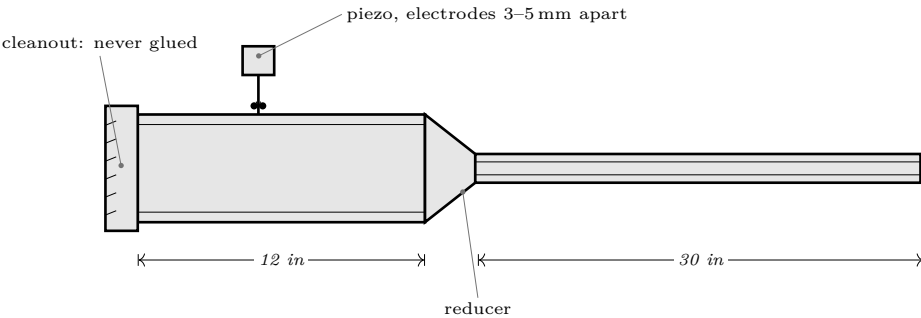
A combustion launcher throws a half-pound potato several hundred feet at speeds that will kill a person. It is not a toy, it is not a firework, and the fact that it is made of plumbing parts changes nothing about what it does downrange. Every rule on the **Safety** sheet applies every time: adult present and in charge, muzzle discipline as strict as a rifle's, a confirmed clear range, eyes and ears on, and never a person, animal, building, road or anything you cannot afford to destroy in the direction it is pointing.

Parts

Part	Typical	Notes
Chamber	3 in Sch 40 pressure PVC, about 12 in long	The volume that holds the fuel and air. Pressure-rated only.
Cleanout	3 in threaded cleanout adapter and plug	The breech. Screws off to load fuel, to vent a misfire, and to wipe residue out. Do not glue the plug.
Reducer	3 in to 1.5 in bushing or reducing coupler	Joins chamber to barrel.
Barrel	1.5 in Sch 40 pressure PVC, about 30 in	Sized so a potato is a firm push fit.
Igniter	BBQ or gas-lighter piezo unit	Two electrodes into the chamber, 3–5 mm apart, roughly a third of the way along.
Electrodes	Two stainless screws with nuts and washers	Through drilled and sealed holes in the chamber wall.
Cement	PVC primer and Sch 40 solvent cement	Both. Primer is not optional; it softens the surface so the weld actually forms.
Fuel	Aerosol hairspray	Cheap, widely available, and hard to over-deliver badly in a single measured burst.
Ammunition	Potatoes slightly larger than the bore	Firm ones. A soft potato disintegrates instead of sealing.

Pressure-rated pipe only. Schedule 40 *pressure* PVC, printed with an NSF-PW rating and a pressure figure. Never DWV, never foam-core, never cellular-core, never thin-wall drain pipe — those are not rated for internal pressure at all and they fail by shattering into fragments rather than splitting. This is the one specification in the whole project that is not negotiable.

Build it



1. **Cut square.** A mitre box or a pipe cutter. An out-of-square cut does not seat fully and leaves a weak joint.
2. **Deburr inside and out.** A knife or sandpaper. A burr inside the barrel scores every potato and a burr inside a socket ruins the weld.
3. **Dry-fit everything.** Check the whole assembly before any cement comes out. Mark each joint with a pencil line across it so you can see if it has rotated home.
4. **Drill the electrode holes** in the chamber, about a third of the way from the breech. Fit the screws with washers and nuts and seal them with epoxy on the outside. Set the internal gap at 3–5 mm and check that the piezo sparks across it before you glue anything.
5. **Prime, then cement.** Primer on both surfaces, cement on both surfaces, push together with a quarter-turn twist, and **hold for thirty seconds** — solvent joints push themselves apart while they set.
6. **Write today's date on the chamber** and leave it **48 hours**. Not four. This is the step everybody shortens and it is the step that matters.

First firing

1. Walk the range. Mark the firing line. Read the checklist on the **Safety** sheet aloud.
2. **Test the spark** with the cleanout open and no fuel: you should hear and see it clearly across the gap.
3. **Load the potato** from the muzzle. Push it in about 6 in with a dowel — firm, sealing, but not so deep that it starts against the reducer.
4. **Open the cleanout, spray a measured one-second burst**, and screw the plug fully home. The adult does this and nobody else.
5. **Shoulder or brace it, muzzle downrange**, everyone behind the line, call the shot, and fire.
6. If it does not fire: **sixty seconds**, muzzle still downrange. Then open the cleanout, fan it out, wait, and start over with a fresh dose. Never add fuel to a failed shot.
7. Between shots, **open the cleanout and fan the chamber** — a chamber full of exhaust has no oxygen in it and the next shot will fail if you do not.

Tuning it

Fuel first	Vary the spray count in half-second steps, three shots each, and measure range. There is a clear peak, and both sides of it are worse. That peak is the stoichiometric ratio you predicted in Lesson 2.
Then the potato	Too loose and gas blows past it; too tight and it will not accelerate. A firm push fit that takes real effort with a dowel is right.
Then the barrel	If you built a spare barrel, compare. Longer gives the pressure more distance to work over, up to the point where friction and falling pressure take over. Lesson 3 predicts roughly where.
Keep a log	Date, fuel dose, barrel, angle, range. Three columns and a pencil. Without it you are guessing, and with it you have a real experiment.
Air it out	Between every shot. This one habit fixes most reliability complaints.

If it does not work

Symptom	Look at
No spark	Piezo faulty, gap too wide, or an electrode loose. Test with the cleanout open, always.
Spark but no bang	Too much fuel, far more often than too little. Vent, fan out, and try a shorter burst. Or the chamber still holds exhaust from the last shot.
Weak pop, potato dribbles out	Under-fuelled, or the potato is not sealing. Check the fit first.
Bang but poor range	Potato too loose, barrel too short, or the potato is soft and disintegrating.
Fires only sometimes	Residue on the electrodes, or you are not airing the chamber between shots.
Any crazing or whitening	Stop. Retire the chamber. <i>see the Safety sheet</i>

What you built

A single-cycle heat engine. Fuel and air burn in a fixed volume, the pressure rises by roughly a factor of three, and that pressure does work on the base of a potato for the length of the barrel. Everything you measured in the four lessons — the pressure of the atmosphere, the fuel-air ratio, the gas law, the projectile's arc — is happening in that fifth of a second, and you can predict the range before you fire it.

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